



SPARKLE: Zero-Shot Discovery of High-Performance, Low-Cost Organic Battery Materials Using Machine Learning



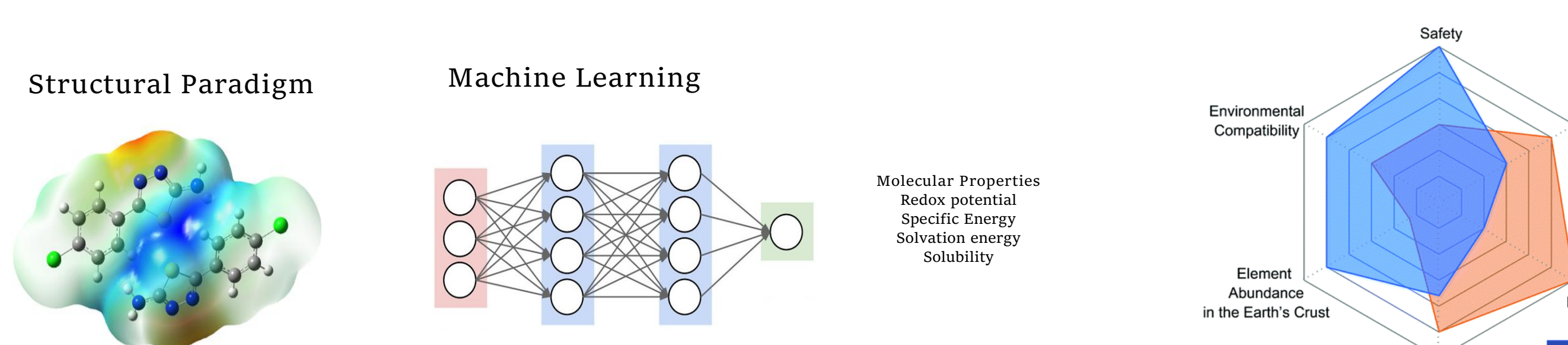
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“Traditional pipelines screen what’s familiar, SPARKLE discovers what’s possible”

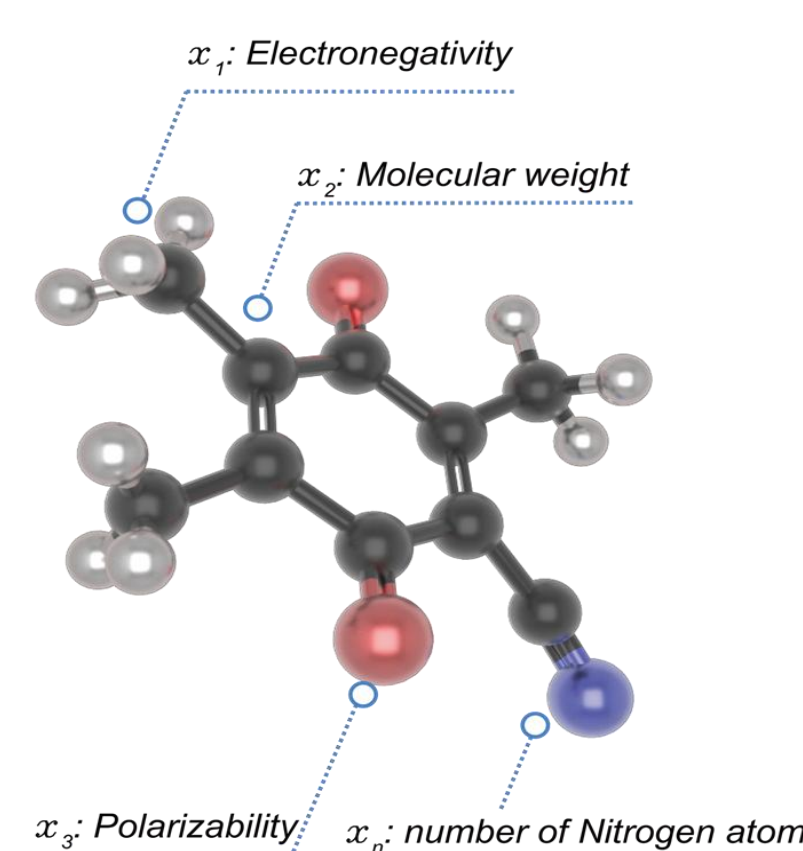
Motivation

- Organic Electrode Materials (OEMs) emerge as a compelling alternative due to their low-cost, tunable electrochemical properties, and environmental sustainability.
- Leveraging Machine Learning (ML) to predict and optimize the electrochemical properties of these materials accelerates the identification of high-performance molecular candidates

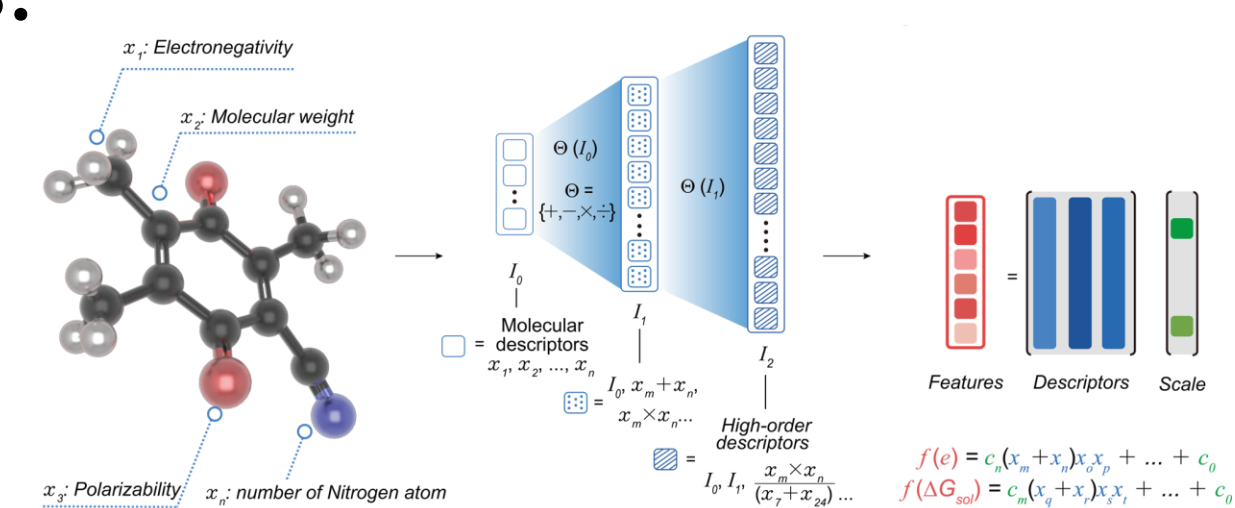


Introduction

- Molecular Descriptors: Numerical representation of the chemical information encoded within a symbolic representation of molecule.

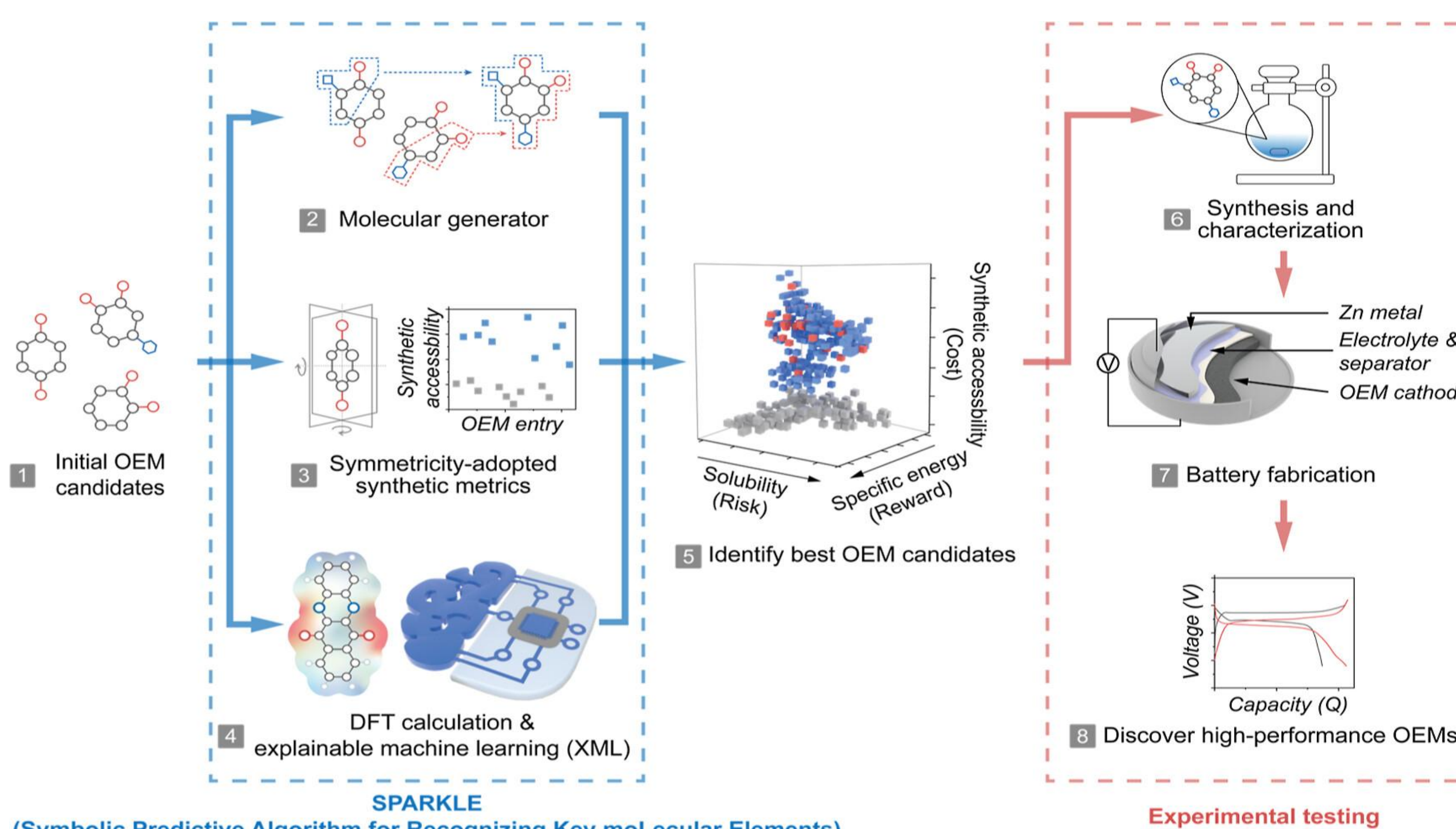


- Traditional ML algorithms leverage molecular descriptors to predict molecular properties, but they often function as black boxes, offering little insight into how predictions are derived or the underlying relationships.
- To address this, we introduce SPARKLE – a novel framework that employs symbolic regression with molecular descriptors, generating interpretable analytical equations that enhance both extrapolation and insight into the underlying relationships.



SPARKLE

- SPARKLE** (Symbolic Predictive Algorithm for Recognizing Key moLecular Elements) is a framework designed to accelerate the design of new molecules for battery application with the desired properties.



- In-house derived Symmetry Adapted Synthetic Accessibility (SASA) is used as cost.

$$SASA = \frac{\text{Total Number of Atoms}}{SCS + \text{Unique Atoms}}$$

*SCS – Synthetic Complexity Score (Neural Network trained on 12 million reactions to predict the score of molecule)

SASA score of ≥ 2.5 as a threshold for easy to synthesize (validated on ~400 molecules by 4 chemists)

$$\text{Specific Energy} = a \left(\frac{x_{BE}}{x_{BP}(x_{MB1} + x_{MB2})} \right) + b$$

Expected, higher electronegativity $\rightarrow$ higher reduction potential

Expected, specific energy inversely proportional to molecular weight

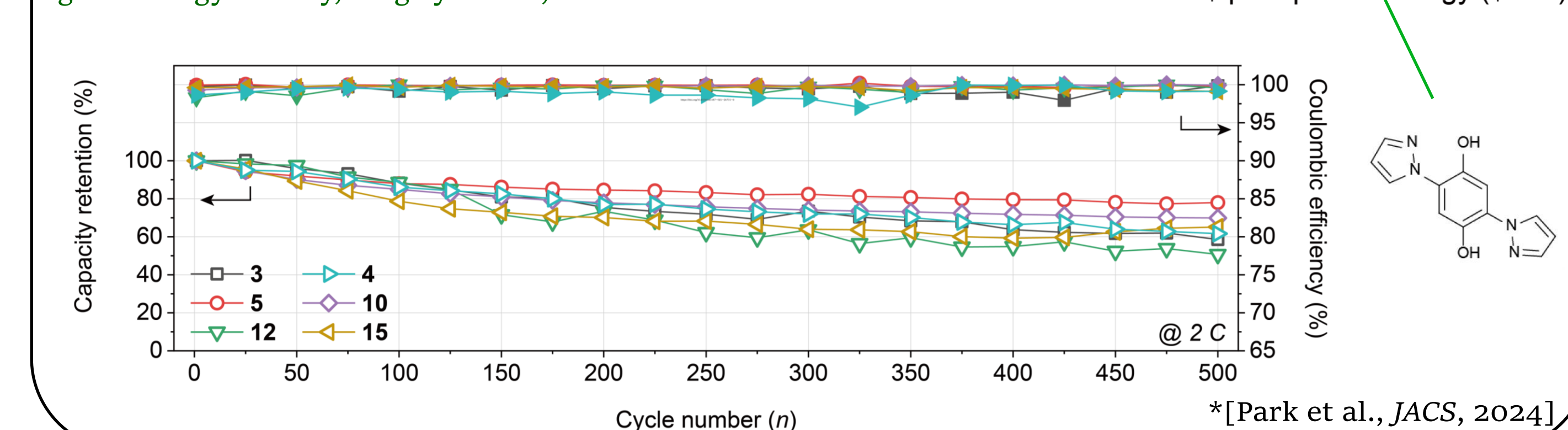
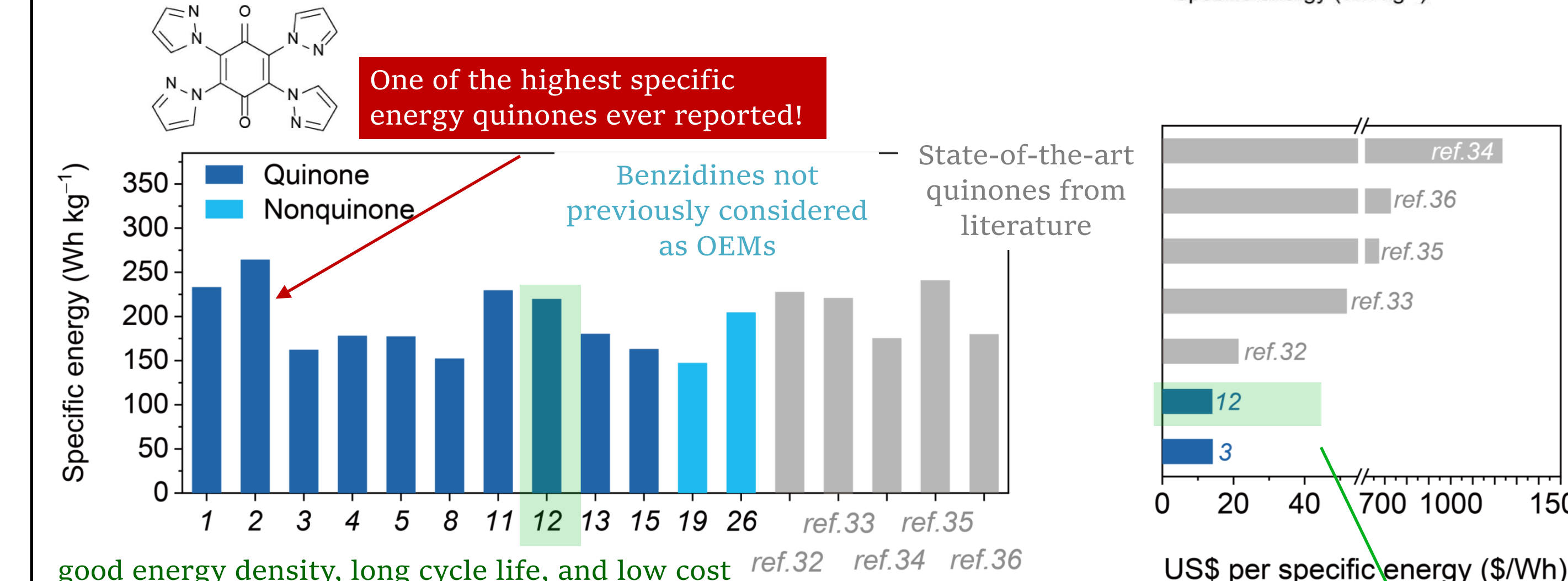
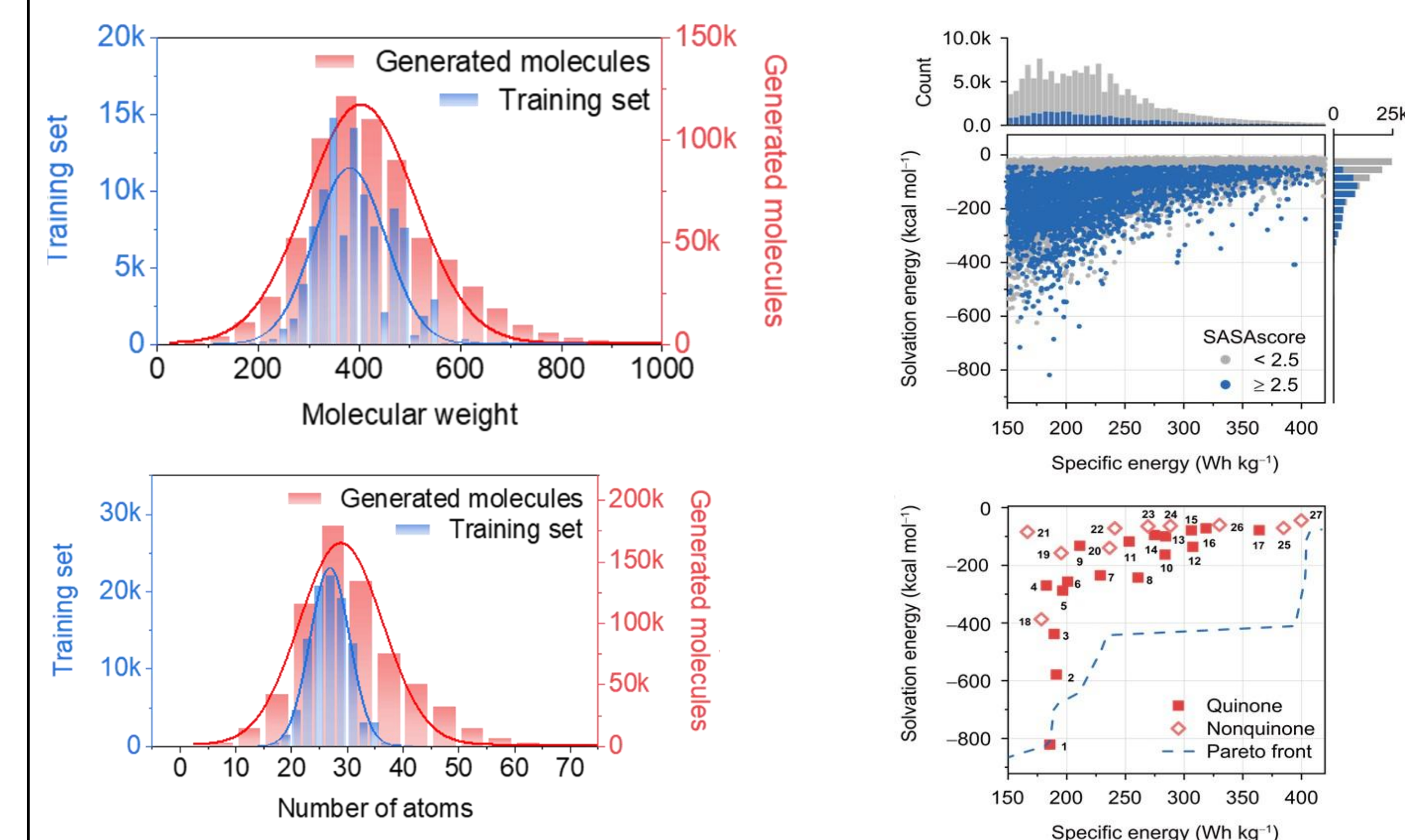
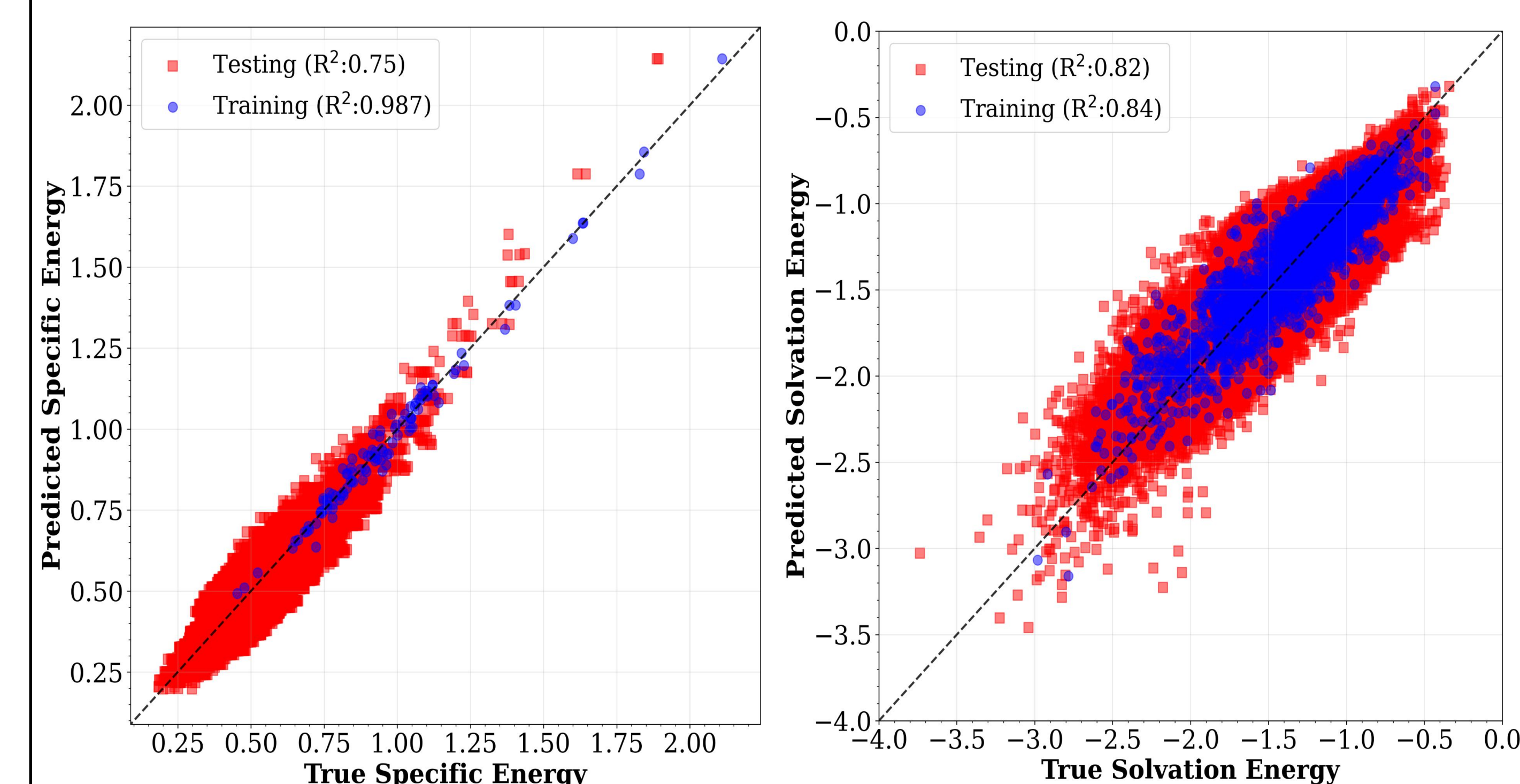
x_{BE} = molecular descriptor related to electronegativity

x_{BP} = molecular descriptor related to polarizability

x_{MB1}, x_{MB2} = molecular descriptors related to molecular weight

Results

Training and Testing evaluation of properties



*[Park et al., JACS, 2024]